

Multispectral Imagery for Wind Turbine (Windmill) Applications

Project Scope Document — with Open-Source Technology Stack and Sample Outputs

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1. Purpose & Objectives

This document defines the scope of a system that uses multispectral imagery (visible, red-edge, near-infrared and optionally SWIR/thermal bands) across the lifecycle of wind turbines: site selection, blade and structural inspection, environmental monitoring, and operations & maintenance (O&M) optimization. The solution is built entirely on open-source software.

- Automate detection of blade defects (erosion, cracks, moisture ingress, lightning damage) from UAV multispectral captures.
- Produce site-analysis products (NDVI, land cover, soil/water masks) from open satellite data (Sentinel-2, Landsat).
- Track vegetation and environmental change around wind farms for permit compliance.
- Deliver georeferenced defect maps, severity classifications and inspection reports through an open-source GIS/web stack.

2. Multispectral Imagery in Wind Applications

Multispectral sensors capture reflectance in discrete bands beyond human vision — typically Blue, Green, Red, Red-Edge and Near-Infrared (400–1000 nm), optionally extended with SWIR and thermal-IR. Because materials such as composite gel coat, moisture, ice, corrosion, vegetation and bare soil each have distinct spectral signatures, band ratios and indices reveal conditions that ordinary RGB imagery misses. The main application areas across the wind energy lifecycle are:

Application Area	What multispectral imagery provides
Blade inspection & damage detection	Cracks, leading-edge erosion, delamination, pitting and coating degradation; NIR/SWIR reveals moisture ingress and bonding defects under the gel coat; lightning-strike burn signatures; ice detection on blades for de-icing decisions.
Thermal & structural health monitoring	Combined with thermal-IR: hotspots in nacelle, gearbox, generator and electrical junctions; internal blade flaws via thermal gradients; corrosion on tower and flanges.
Site selection & pre-construction survey	Land cover / land use classification from Sentinel-2 or Landsat; soil moisture and terrain indicators for foundations and access roads; NDVI/EVI ecological baselines; surface roughness input to wind-flow modelling.
Environmental & ecological monitoring	Vegetation-health change detection around the farm; bird/bat habitat mapping for curtailment planning; erosion and drainage monitoring; periodic compliance reports.
Operations & maintenance optimization	Repeatable UAV fleet-wide blade scans with minimal downtime; spectral trend analysis for predictive maintenance; digital-twin integration; rapid post-storm triage.
Wake & vegetation studies	Vegetation stress patterns downstream of turbines (microclimate/wake effects); agricultural coexistence studies on leased land.
Offshore-specific applications	Marine growth / biofouling on foundations; coating breakdown in the splash zone; sediment-plume monitoring during construction (phase 2).

3. In-Scope Work Packages

WP	Work Package	Description
WP1	Data acquisition	UAV multispectral flights (blade & tower); ingest of Sentinel-2/Landsat scenes for site & environment products.
WP2	Preprocessing	Radiometric calibration, band alignment/registration, orthorectification, mosaicking, reflectance conversion.
WP3	Index & analytics engine	NDVI/NDRE/EVI computation, anomaly masks, change detection between inspection epochs.
WP4	Defect detection (ML)	CNN / classical CV models for crack, erosion, moisture and lightning-damage classification with severity grading.
WP5	GIS & reporting	Georeferenced defect layers, web map dashboard, automated PDF inspection reports.
WP6	O&M integration	Export to CMMS/digital-twin via open APIs; trend analysis across inspection cycles.

Out of scope (phase 1): SCADA data fusion, offshore marine-growth monitoring, real-time onboard (edge) inference, LiDAR integration.

4. Open-Source Software Stack

Layer	Tool (License)	Role in this project
Satellite data access	ESA Sentinel-2 / USGS Landsat (free & open data); sentinelsat, pystac-client (GPL/Apache)	Free multispectral scenes and API download for site & environmental products.
Photogrammetry / stitching	OpenDroneMap / WebODM (AGPL-3.0)	Orthomosaics and 3D reconstruction from UAV multispectral flights, incl. band alignment for MicaSense-class sensors.
Raster / geospatial core	GDAL (MIT), Rasterio (BSD), rioxarray (Apache-2.0)	Reading, reprojecting, calibrating and writing multispectral rasters.
Scientific computing	NumPy, SciPy, xarray (BSD)	Band math: NDVI, NDRE, EVI, custom anomaly indices.
Image processing / CV	OpenCV (Apache-2.0), scikit-image (BSD)	Denoising, thresholding, morphology, contour extraction for defect masks.
Machine learning	PyTorch (BSD) or TensorFlow (Apache-2.0); scikit-learn (BSD); Ultralytics YOLO (AGPL-3.0)	Defect detection/classification models; random-forest land-cover classification.
Semantic segmentation	segmentation-models-pytorch (MIT)	U-Net/DeepLab blade-surface defect segmentation.
GIS desktop	QGIS (GPL-2.0) + Semi-Automatic Classification Plugin	Visual QA, manual labelling, land-cover classification workflows.
Spatial database	PostGIS on PostgreSQL (GPL/PostgreSQL licenses)	Storage of defect geometries, turbine assets, inspection epochs.
Web mapping / dashboard	GeoServer (GPL-2.0), Leaflet (BSD), Streamlit (Apache-2.0)	Serving tiles and interactive inspection dashboards.
Visualization / reports	Matplotlib (PSF-based), ReportLab (BSD)	Figures and automated PDF inspection reports (this document was produced with them).
Workflow orchestration	Apache Airflow (Apache-2.0)	Scheduled ingestion, processing and report pipelines.

5. Drone (UAV) Platform & Payload Sensor Requirements

5.1 Minimum UAV platform requirements

Requirement	Minimum specification	Why it matters
Airframe type	Multirotor (quadcopter) with stabilized 3-axis gimbal	Hovering and slow orbits around blades; steady captures in gusts.
Payload capacity	≥ 350 g (integrated sensor) or ≥ 900 g (M300/M350-class for external payloads)	Must carry the multispectral sensor plus irradiance (DLS) module.
Flight endurance	≥ 25 –30 min per battery (with payload)	One full blade (3 sides) or one site grid per battery swap.
Positioning	RTK/PPK GNSS, 1–3 cm accuracy (base station or NTRIP)	Repeatable, georeferenced captures for epoch-to-epoch change detection.
Wind tolerance	≥ 12 m/s sustained	Wind farm sites are windy by definition; inspection windows are short.
Obstacle sensing	Omnidirectional avoidance + safety geofence	Flying within 5–15 m of blades and tower.
Mission software	Waypoint / orbit automation with terrain follow (open protocols: MAVLink preferred)	Repeatable automated inspection paths; integration with OpenDroneMap outputs.
Regulatory	Registered platform, remote-ID; pilot licensed per DGCA/FAA/EASA class	Legal operation near tall structures.

5.2 Payload sensor options (named) and minimum sensor specification

Payload sensor (Name)	Bands	Resolution / GSD	Notes
DJI Mavic 3M (Mavic 3 Multispectral) — integrated	RGB + 4 MS bands (G 560, R 650, RE 730, NIR 860 nm)	5 MP MS; ~ 5 cm/px @ 60 m	Minimum-budget entry; RTK module included; sunlight sensor on top.
MicaSense RedEdge-P (external payload)	5 MS bands (B, G, R, RedEdge, NIR) + panchro	1.6 MP/band; ~ 2 cm/px @ 60 m (pan-sharpened)	Industry standard for inspection; global shutter; DLS 2 irradiance sensor.
MicaSense Altum-PT (external payload)	5 MS bands + panchro + thermal LWIR	3.2 MP/band MS; 320x256 thermal	Adds thermal for electrical/structural hotspots; heavier (577 g).
Sentera 6X (external payload)	5 MS bands + RGB	3.2 MP/band	Alternative vendor; global shutter; open SDK.
Parrot Sequoia+ (legacy option)	4 MS bands + RGB	1.2 MP/band	Budget/legacy; acceptable for vegetation products only, not fine blade defects.

Minimum sensor specification for this project: ≥ 5 spectral bands including Red-Edge and NIR (400–1000 nm); global shutter; ≥ 1.2 MP per band; downwelling light/irradiance sensor (DLS) and calibrated reflectance panel for radiometric correction; GPS-tagged 16-bit TIFF output. **Minimum viable configuration:** DJI Mavic 3M (all-in-one, blade + site products). **Recommended full configuration:** DJI Matrice 350 RTK carrying MicaSense Altum-PT (multispectral + thermal in one flight).

6. Required Datasets & File Formats

The table below lists every dataset the project consumes or produces, with its source, file format and extension.

6.1 Input datasets

Dataset (Name)	Source / Sensor	Format	Extension(s)
UAV multispectral blade/tower captures	MicaSense RedEdge-MX / Altum-PT, DJI Mavic 3M (5–7 bands)	16-bit GeoTIFF per band; raw sensor capture	.tif / .tiff, .dng

Dataset (Name)	Source / Sensor	Format	Extension(s)
Radiometric calibration panel images	Same UAV sensor, pre/post-flight panel shots	16-bit TIFF + panel reflectance sheet	.tif, .csv
Sentinel-2 L2A scenes (site & environment)	ESA Copernicus Open Access / AWS Open Data	SAFE archive with JPEG2000 band files, XML metadata	.SAFE (folder), .jp2, .xml
Landsat 8/9 Collection-2 Level-2 scenes	USGS EarthExplorer	Cloud-Optimized GeoTIFF + MTL metadata	.tif, .txt / .xml (MTL)
Digital elevation model (terrain)	Copernicus DEM GLO-30 / SRTM	GeoTIFF raster	.tif
Turbine asset registry & farm boundary	Operator GIS / survey	Vector layers (points, polygons)	.gpkg, .shp (+.dbf/.shx/.prj), .geojson
UAV flight logs & telemetry	Flight controller export	Tabular log / GPS track	.csv, .log, .gpx
Weather & wind data (QA filtering)	ERA5 reanalysis / met mast	NetCDF grids or tables	.nc, .csv
Public blade-defect training data (e.g. DTU 'Nordtank' turbine inspection images)	DTU Data / Mendeley open repositories	Annotated RGB/NIR images	.jpg, .png

6.2 Labels, models and intermediate data

Dataset (Name)	Produced by	Format	Extension(s)
Defect annotation labels	Manual labelling (QGIS / CVAT / Label Studio)	COCO JSON, YOLO text labels, mask rasters	.json, .txt, .png
Calibrated reflectance orthomosaics	OpenDroneMap / WebODM (WP2)	Multi-band Cloud-Optimized GeoTIFF	.tif
Index rasters (NDVI / NDRE / EVI)	NumPy / xarray pipeline (WP3)	Single-band GeoTIFF; gridded stacks	.tif, .nc
Trained ML model weights	PyTorch / YOLO training (WP4)	Serialized weights; portable inference model	.pt / .pth, .onnx
Anomaly / defect masks	OpenCV + model inference (WP4)	Binary/label mask rasters	.png, .tif

6.3 Output / deliverable datasets

Dataset (Name)	Consumed by	Format	Extension(s)
Defect detection layer (geometry + severity class)	QGIS, web dashboard, CMMS	GeoPackage / GeoJSON vector layers; PostGIS tables	.gpkg, .geojson (+ SQL)
Land-cover & change-detection maps	Environmental compliance reporting	GeoTIFF raster + style file	.tif, .qml / .sld
Inspection report per campaign	O&M engineers, asset owners	PDF report; tabular defect register	.pdf, .csv / .xlsx
Web map tiles & dashboard layers	GeoServer / Leaflet / Streamlit	XYZ/WMTS tile cache, MBTiles	.png (tiles), .mbtiles
Pipeline metadata & lineage	Airflow / audit	Run logs, STAC item metadata	.json, .log

Conventions: all rasters are EPSG-referenced GeoTIFFs (Cloud-Optimized where served over the web); vectors default to GeoPackage (.gpkg) to avoid multi-file shapefile handling; naming pattern `<site>_<turbine>_<date>_<product>.<ext>`, e.g. `WF01_T07_20260723_ndvi.tif`.

7. Sample Outputs (Synthetic Demonstration Images)

The images below are synthetic samples generated with NumPy + Matplotlib to illustrate the expected products. Production outputs will use real UAV and Sentinel-2 data processed through the stack in Section 4 over the datasets in Section 5.

7.1 Site analysis product — multispectral bands and NDVI

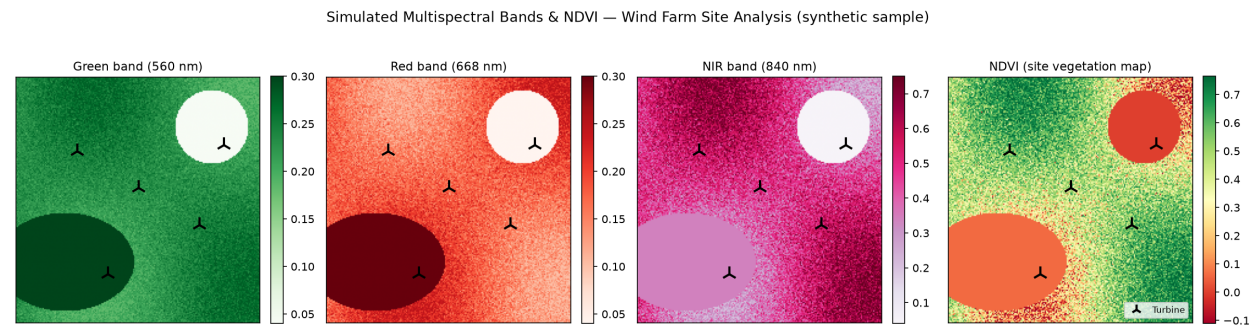


Figure 1 — Simulated green/red/NIR reflectance and derived NDVI over a wind farm site. Turbine symbols mark asset locations; low-NDVI areas (water body, bare soil) are automatically excluded from vegetation-compliance monitoring.

7.2 Blade inspection product — NIR capture and automated defect mask

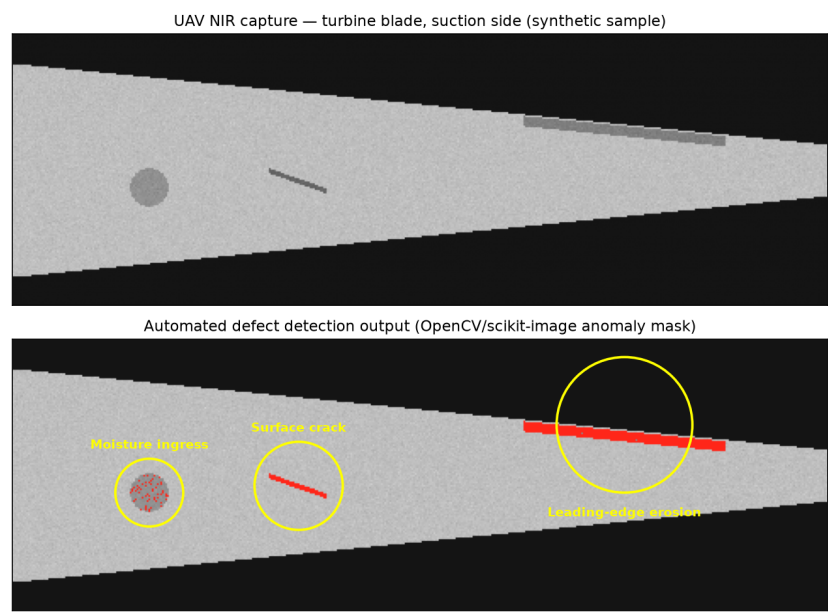


Figure 2 — Simulated UAV NIR image of a blade (top) and the automated anomaly detection overlay (bottom): leading-edge erosion, a surface crack and a moisture-ingress zone are flagged in red with yellow region-of-interest markers.

8. Processing Pipeline

Step	Stage	Open-source tools
1	Acquire — UAV flight plans / satellite scene download	OpenDroneMap ground control, sentinelsat/pystac-client
2	Calibrate — radiometric correction, reflectance panels	OpenDroneMap, Rasterio, NumPy
3	Register & mosaic — band alignment, orthomosaic	OpenDroneMap/WebODM, GDAL
4	Analyze — indices (NDVI/NDRE), anomaly masks	NumPy, xarray, OpenCV, scikit-image
5	Detect — ML defect classification & severity grading	PyTorch, segmentation-models-pytorch, YOLO
6	Store & serve — geodata layers, tiles, dashboard	PostGIS, GeoServer, Leaflet, Streamlit

Step	Stage	Open-source tools
7	Report — automated inspection PDFs, trend charts	Matplotlib, ReportLab, Airflow (scheduling)

9. Workflow & Methodology Adopted

Delivery follows an **agile, phased methodology**: the analytics components (WP3–WP4) use a **CRISP-DM adapted loop** (business understanding → data understanding → preparation → modelling → evaluation → deployment), while field operations follow a fixed, auditable inspection workflow per campaign. Every stage has an entry criterion, an exit (QA) gate and a named artefact, so a campaign is traceable end-to-end.

9.1 End-to-end inspection workflow (per campaign)

#	Workflow step	Method / procedure	Exit gate → artefact
1	Campaign planning	Select turbines/site; define flight plans (orbit per blade side, lawn-mower grid for site); check weather window and permits.	Approved flight plan → mission file (.kmz/.plan)
2	Pre-flight calibration	Capture reflectance panel; verify DLS; RTK fix confirmed; sensor self-test.	Calibration checklist passed → panel images (.tif)
3	Data acquisition	Automated waypoint flight; 75–85% overlap; GSD per product spec; turbine stopped & locked for blade scans.	Coverage check on-site → raw band captures (.tif)
4	Ingest & preprocessing	Radiometric correction, band alignment, orthomosaic (OpenDroneMap); reflectance conversion.	QA-1 data gate → calibrated orthomosaic (.tif)
5	Analytics & detection	Index computation (NDVI/NDRE); ML inference for defect classes; severity grading rules.	QA-2 model gate → defect layer (.gpkg)
6	Human review (QA)	Analyst reviews flagged defects in QGIS/dashboard; confirms, edits or rejects each detection.	QA-3 review gate → verified defect register (.csv)
7	Reporting & handover	Automated PDF report; dashboard update; export to CMMS; archive with lineage metadata.	AC sign-off → campaign report (.pdf)

9.2 ML model development methodology (CRISP-DM adapted)

- Data understanding — exploratory band statistics on pilot flights; label taxonomy fixed with O&M engineers (crack, erosion, moisture, lightning, other).
- Preparation — tiling, augmentation, band-stacking; 70/15/15 train/validation/test split, stratified by turbine and defect class; test set frozen before training.
- Modelling — baseline classical CV first (thresholds/morphology), then U-Net segmentation and YOLO detection; experiments tracked in MLflow (open source).
- Evaluation — metrics on the frozen test set only (Section 10 AC thresholds); error review board with a blade engineer before any release.
- Deployment — versioned ONNX export; inference in the Airflow pipeline; shadow-mode run for one campaign before replacing the incumbent model.
- Monitoring — drift checks on band histograms and detection rates each campaign; retraining triggered when AC metrics degrade.

10. Quality Assurance (QA) & Acceptance Criteria (AC) Report

QA is enforced at three gates in the workflow (Section 9.1); a campaign is accepted only when every AC in the table below is met. Each campaign produces a QA/AC report recording the measured value against each criterion, signed by the analyst and the client's O&M representative.

10.1 QA gates and checks

Gate	Scope	Checks performed
QA-1 — Data quality	Raw captures & orthomosaic	Image sharpness (blur metric), exposure/histogram sanity, band-to-band alignment error, forward/side overlap achieved, GSD achieved, radiometric panel deviation, GNSS fix quality, coverage completeness vs flight plan.
QA-2 — Model output	Indices & ML detections	Index value ranges plausible (NDVI within [-1, 1], no saturation), detection confidence distribution reviewed, per-class counts vs historical norms, no empty/failed tiles, model version and checksum logged.
QA-3 — Human review	Defect register	100% of Severity 3+ detections reviewed by analyst; >=20% random sample of lower severities; inter-analyst agreement spot-checks; final register free of duplicates and geometry errors.

10.2 Acceptance criteria (AC)

ID	Acceptance criterion	Threshold	Verification method
AC-1	Blade coverage per inspected turbine	>= 95% of blade surface imaged (all 3 blades, both sides + leading/trailing edge)	Coverage map vs blade CAD footprint
AC-2	Ground sample distance (blade products)	<= 3 mm/px on blade surfaces	EXIF/flight-log distance & sensor model
AC-3	Ground sample distance (site products)	<= 10 cm/px	Orthomosaic metadata
AC-4	Georeferencing accuracy	<= 10 cm RMSE horizontal (RTK), <= 2 px co-registration between epochs	Check-point residuals report
AC-5	Radiometric calibration	Panel-derived reflectance within $\pm 5\%$ of certified values per band	Calibration report per flight
AC-6	Defect detection recall (Severity 3+)	>= 90% on frozen test set and field-verified sample	Confusion matrix in QA/AC report
AC-7	Defect detection precision (all classes)	>= 80% (false positives <= 20%)	Confusion matrix in QA/AC report
AC-8	Human review completion	100% Severity 3+, >= 20% sample of lower severities	Review log export
AC-9	Report turnaround	Draft report <= 3 working days after last flight; final <= 5 after review	Timestamps in Airflow lineage
AC-10	Data completeness & lineage	All deliverables present, named per convention, with STAC metadata and checksums	Automated manifest validation

QA/AC report contents (per campaign): campaign summary (site, turbines, dates, crew, weather), flight log table, QA-1/2/3 gate results with measured values, AC table with pass/fail per criterion, confusion matrix and metric plots, deviations & waivers with justification, and sign-off block (analyst, QA lead, client representative). The report is generated automatically by the ReportLab pipeline and archived alongside the campaign data.

11. Deliverables

- D1 — Calibrated multispectral orthomosaics per turbine/site (GeoTIFF).
- D2 — Defect detection layer with severity classes (GeoPackage/PostGIS).
- D3 — NDVI / land-cover / change-detection maps for environmental compliance.
- D4 — Web dashboard (Leaflet/Streamlit) with per-turbine inspection history.
- D5 — Automated PDF inspection reports with QA/AC section (per flight campaign).

- D6 — Reproducible open-source pipeline (containerized, Apache Airflow DAGs).

12. Standards, Assumptions & Constraints

- Inspection practice aligned with IEC 61400 series and DNV-GL blade inspection guidance.
- UAV operations subject to local aviation (e.g. DGCA/FAA/EASA) rules; flights only in permitted wind conditions.
- Multispectral sensor with at least 5 bands (B, G, R, RedEdge, NIR); SWIR/thermal optional add-ons.
- All software components are OSI-approved open source; AGPL components (WebODM, YOLO) used as services to keep licensing obligations contained.
- Sample images in this document are synthetic and for illustration only.